# Modelling & Simulation: Assignment 4 (option 1)
# **Variable Stiffness Actuator**

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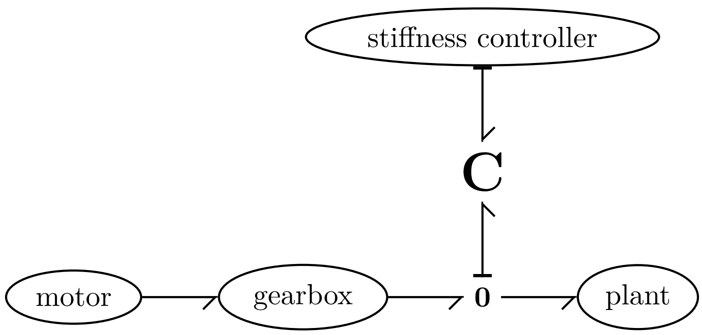
## Introduction

In assignment 3, you modelled a series elastic actuator (SEA). While this system has some benefits in efficiency and safety or robustness, the system is not very flexible. That is to say, the elastic element is very flexible of course, but it has a fixed stiffness. This means that the system (drive train) can be optimised for a certain load and motion profile, but outside of this range a lot of efficiency may be lost:

- If the system has a varying load, for example because a robot arm is picking up heavy objects, then the bandwidth of the system will differ a lot between the "carrying object" and "gripper empty" states.
- If the arm is controlled using impedance control, this will be very efficient if the intended stiffness is exactly equal to the stiffness of the elastic element. For a higher or lower apparent stiffness, the motor will have to rotate a lot to "wind and unwind" the spring.
- If the SEA is connected to a robot arm, the apparent output stiffness will depend on the arm configuration (angle): the kinematics of the robot arm will modulate the connection between elastic element and end-effector.

A possible solution is the VSA: the Variable Stiffness Actuator. This is a SEA in wich the (apparent) stiffness of the elastic element can be changed. There are many ways in which to realise this: using variable lever arms, nonlinear stiffness, continuously variable transmission, et cetera. Looking at the system from the outside, however—focussing on the port behaviour—it can be modelled quite simply:




This two-port **C**-element has the regular mechanical port with variables $\langle F, v\rangle$ or $\langle \tau, \omega\rangle$; and a stiffness-changing port $\langle e_K, \dot{K}\rangle$. $\dot{K}$ is the rate of change of stiffness and $e_K$ the associated effort.

*Assignment details on second page.*

## Assignment

**Model a VSA by adding a stiffness-changing port to the spring of SEA in assignment 3. Design a stiffness-changing controller and show what changes in the energy flows are with respect to a SEA.**

The two-port **C**-element is not hard to model, because its port behaviour—resulting from the actual spring and the stiffness-changing mechanism—is fully described by its energy function:

$$H_c(q, K) = \frac{1}{2}Kq^2 \tag{1}$$

$$e_q = \frac{\partial H_c}{\partial q} = Kq \tag{2}$$

$$e_K = \frac{\partial H_c}{\partial K} = \frac{q^2}{2} \tag{3}$$

### Some things to consider

1. After replacing the **C**-element in the SEA of assignment 3, you have to come up with a controller that changes the stiffness, i.e. that controls $\dot{K}$.
2. Is there a clear objective or optimal $K$ for torque control—step 2 of assignment 3?
3. How is that for impedance control?
4. What is the energy cost of changing the stiffness of a VSA? Is it possible to minimise this somehow?

### Additional questions you could try to answer are:

- The mechanism in a VSA is usually controlled by a (small) electric motor, and losses occur in the motor and the stiffness-changing mechanism. How could you model those losses in this simplified VSA model?
- Does this change the efficiency or energy flows a lot?
- When does a VSA have a large benefit over an SEA?

Suggested material:


- L. C. Visser, R. Carloni, and S. Stramigioli, "Variable stiffness actuators: A port-based analysis and a comparison of energy efficiency", IEEE International Conference on Robotics and Automation (ICRA), pp. 3279–3284, 2010.